

A Guided Tour of the Forecasting Experiment

From Belgian quarter-hourly load to spectral structure, leakage-safe forecasts, uncertainty bands and untouched confirmation

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Five-minute summary

Measured: Belgian quarter-hourly day-ahead load forecasting using only temporal structure and recent load, with an 18:00 D-1 information boundary.

Untouched confirmation: 1 February-31 December 2025, 334 civil days; point comparisons use a common sample of 32,060 quarter-hours.

MAE: D-7 438.9 MW; D-7 + identical recent level 423.0 MW; frozen Fourier/calendar + recent level 315.4 MW; Elia 277.7 MW.

Confirmed attribution: the frozen model improves MAE by 25.4% versus the information-matched persistence control, 95% seven-day block-bootstrap interval 18.8%-32.0%.

Elia remains better. That is part of the finding: a transparent load-history-only model recovers substantial signal, but not the full operational advantage.

Three minutes: read this box and §11. Fifteen minutes: add §5, §6 and §7.

The story before the statistics

Electricity demand has an obvious rhythm. Offices open, people cook, factories work, weekends differ from weekdays, and the same calendar keeps returning. That makes load look predictable. But a repeating pattern is not the same thing as a useful day-ahead forecast.

This study asks a deliberately narrow question: how much of Belgian quarter-hourly electricity demand can be predicted one day ahead from temporal structure and recent load information alone? The professional Elia day-ahead forecast is used as an operational reference, not as training data and not as a like-for-like competitor.

The central discipline. At 18:00 Belgian civil time on D-1, the model must behave as if the future has not happened. Every feature, residual and benchmark is audited against that information boundary.

The most important result is not that a simple model beats Elia - it does not. It is that the experiment separates three questions that are often mixed together: what recurring structure contributes, what recent level information contributes, and what uncertainty can honestly be claimed after the specification is frozen.

Roadmap of the argument

Section	Question
§1 Data contract	What is known at the 18:00 D-1 forecast origin?
§2 Spectral anatomy	What periodic structure is visible in Belgian load?
§3 Baselines	How hard is it to beat honest civil-time persistence?
§4 Structural model	What exactly is the sine/cosine regression basis?
§5 Diagnosis and correction	Is the miss mainly shape or day-level positioning?
§6 Freeze	What was fixed before untouched outcomes were opened?
§7 Point confirmation	What point-forecast result survives 2025?
§8 Probabilistic forecast	Do empirical P10-P90 intervals quantify uncertainty?
§9 Conditional calibration	Where does calibration look weak, without overclaiming?
§10 Inference	How uncertain are headline comparisons once days are dependent?
§11 Conclusions	What did the experiment actually establish?
§12 Next decision problem	Does forecast accuracy translate into battery value?

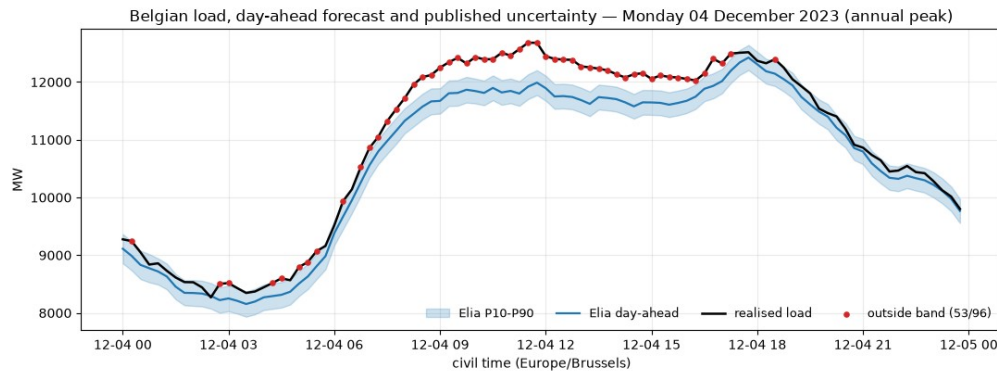
Reproducibility note. Guided Tour § numbers and notebook § numbers are separate. References below give both the notebook section and its explicit CELL address in `probabilistic_forecasting.ipynb`.

1. Start with the forecast contract

Notebook reference: §0-§3, CELL 02-07 (forecast contract, data preparation, Elia audit and load/forecast figures).

The raw material is Elia Open Data dataset ods001: measured Belgian total load together with Elia point and quantile forecasts. The analysis is performed in Europe/Brussels civil time rather than by blindly shifting rows in UTC. The figure below shows one complete civil day - 96 quarter-hours - together with the operational Elia forecast used as the study's reference.

Civil day	Quarter-hour slots	Why
Ordinary day	96	24 hours × 4
Spring-forward day	92	02:00-02:45 does not exist
Fall-back day	100	02:00-02:45 occurs twice



What is being forecast: Belgian quarter-hourly load on the 2023 annual-peak day, with Elia's day-ahead point forecast and published P10-P90 interval.

That detail matters immediately. A D-7 forecast for 6 April 2025 needs the corresponding civil times on 30 March, the spring-forward Sunday. Four source times do not exist. The study does not invent them. Those four target slots are removed from every point model when the common-sample comparison is made.

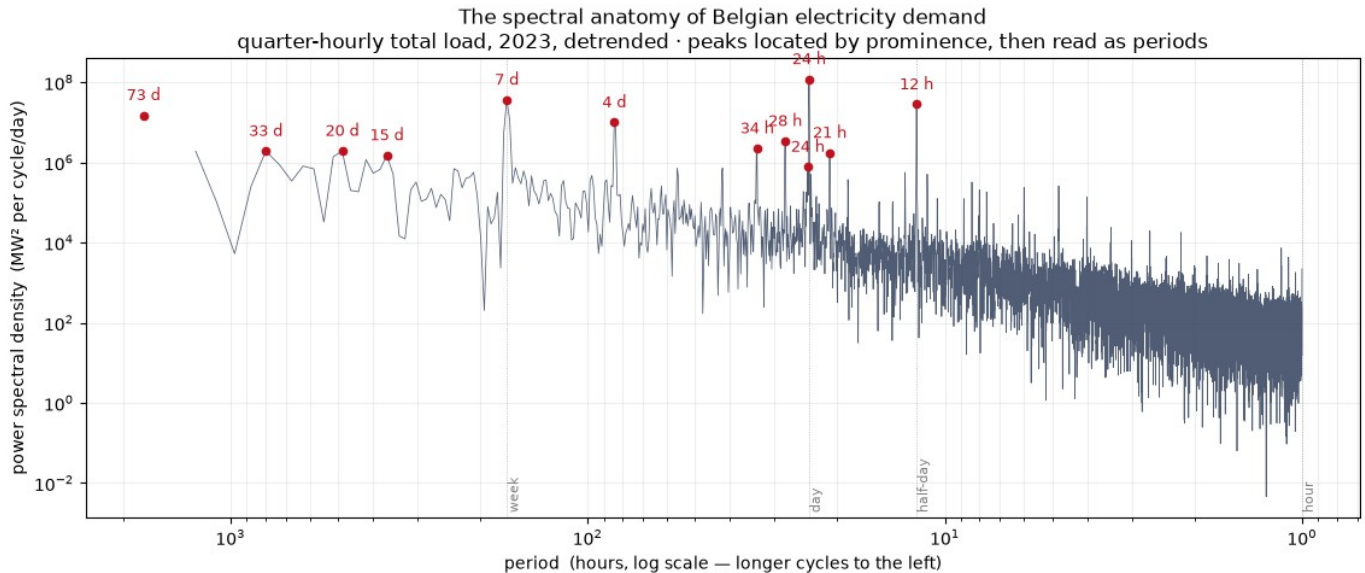
Why civil time matters. Electricity use follows clocks and calendars. A mechanically shifted UTC series can turn a daylight-saving transition into a modelling artefact.

The ODS001 field metadata labels the operational reference as a “Day-ahead 6PM forecast”. The study therefore fixes its own origin at 18:00 Europe/Brussels on D-1. This wording is intentionally tied to the dataset field metadata; other Elia pages have used different publication times.

2. Spectral anatomy: repetition is visible, but not yet predictive

Notebook reference: §4, CELL 08-11 (uniform-grid preparation, periodogram, spectral figure and interpretation guard).

Before fitting the forecasting model, the 2023 realised-load series is inspected with a periodogram: a chart showing how much of the series' variation is associated with each repetition period. Peaks are detected by prominence - how strongly a peak stands above its local neighbourhood - and only then translated into periods. The aim is to let the data reveal its recurring rhythms before we fit a regression.



The spectral-anatomy figure from the notebook's spectral-analysis section. Red points mark prominent components retained for presentation; the 0.1% share rule is a descriptive relevance threshold, not a significance test.

The dominant resolved components include the 24-hour cycle, the weekly component near 7 days and the 12-hour harmonic. The weekly cycle does not land exactly on the discrete frequency grid because one civil year does not contain an integer number of weeks, so some of its power is spread across neighbouring frequencies.

Three missing actual-load observations are filled for this descriptive diagnostic because the transform requires a uniform grid. The selected component set survives an independent weekly-gap-fill check. Recomputing without the Hann window - a standard taper used to reduce edge effects - does not reproduce exactly the same set. That discrepancy is reported rather than promoted to a successful confirmation; power spreading across neighbouring frequencies is a plausible explanation.

What the spectrum licenses - and what it does not. It establishes that recurring daily and weekly structure is genuinely present in realised Belgian load. It does not license us to pick whichever full-year peaks make a forecasting model look best, because that would use the evaluation year to choose the model.

That distinction leads directly to the regression basis. The forecasting frequencies are not hand-picked from this plot. They are generated mechanically from the civil calendar: the first four harmonics of a day and the first three harmonics of a week. The full-year spectrum is an independent descriptive sanity check that those calendar periodicities are real.

3. Baselines: make the simple competitors honest

Notebook reference: §5, CELL 12-16 (civil-time persistence, origin audit, common-sample scoring and baseline error anatomy).

A forecasting model should first beat something that an operator could actually have produced at the stated origin. The main accuracy measure is mean absolute error (MAE): the average absolute difference between realised and forecast load, in MW. Two persistence ideas are useful: use the same civil time yesterday, or use the same civil time one week ago.

Previous-day persistence looks natural, but at an 18:00 D-1 origin part of the target day is only six hours away and many D-1 source observations for later target slots have not happened yet. It is therefore partly informed and is kept only as a descriptive comparator. D-7 is the clean operational baseline.

2023 common sample	MAE (MW, 2023)	Interpretation
Previous day D-1	482.3	Partly informed at the chosen origin
Previous week D-7	433.8	Clean operational persistence
Elia day-ahead	259.9	Professional operational reference

Weekly persistence beats daily persistence overall because the weekly calendar matters. Holidays are its obvious failure mode: the same weekday one week earlier is often an ordinary working day.

2023 sample note. The 2023 figures are recomputed within each section on the common sample available to the models compared there. They are therefore directly comparable within a table, but small differences between sections (for example D-7 = 433.8 versus 418.7 MW) reflect different common samples rather than inconsistent arithmetic.

4. The structural forecast: intentionally small

Notebook reference: §6, CELL 17-22 (Fourier/calendar fits, temporal-integrity audit, error decomposition and basis audit).

Underneath the terminology, this is an ordinary linear regression. Each block in the table below becomes a block of predictor columns in the regression. The corresponding coefficients appear in the equation: β_0 and β_N for the level terms, a and b for the daily sine/cosine terms, and c and d for the weekly sine/cosine terms.

$$\hat{y}_t = \beta_0 + \beta_N N_t + \sum_{k=1}^4 [a_{k, c(t)} \sin(2\pi k d_t) + b_{k, c(t)} \cos(2\pi k d_t)] + \sum_{j=1}^3 [c_j \sin(2\pi j w_t) + d_j \cos(2\pi j w_t)]$$

$$d_t = \text{quarter-of-day}/96, \quad w_t = (\text{96 day-of-week} + \text{quarter-of-day})/672$$

The frozen Fourier/calendar point model. β_0 is the intercept; N_t equals 1 on a non-working day and 0 otherwise; β_N is the fitted non-working-day level shift. $c(t)$ identifies the day class (working or non-working). The a/b coefficients multiply the daily sine/cosine columns; c/d multiply the weekly sine/cosine columns.

Every sine needs its cosine partner. Together they let the fitted cycle choose both amplitude and phase. For example, the 24-hour pair can represent a daily wave whose peak occurs at any clock time; the coefficients decide how large the wave is and where it sits. Ordinary least squares estimates all coefficients simultaneously from the historical load observations.

Basis block	Terms supplied to OLS	Selection rule
Daily	$k = 1..4 \rightarrow 24, 12, 8, 6 \text{ h}; \sin + \cos$	First four harmonics of the civil day
Day class	Daily block split: working / non-working	One pre-specified shape interaction
Weekly	$j = 1..3 \rightarrow 168, 84, 56 \text{ h}; \sin + \cos$	First three harmonics of the 7-day calendar
Level	Intercept + non-working indicator	Baseline and non-working level shift
Annual	None	Not identifiable near the start from available history

The resulting design matrix has 24 columns: one intercept, one non-working level term, sixteen daily sine/cosine columns (eight for working days and eight for non-working days), and six weekly sine/cosine columns. For each target day the model is re-fitted by ordinary least squares on a rolling 56-day history ending at the 18:00 D-1 forecast origin.

Why this is not feature shopping. The basis is fixed by a calendar rule, not selected because a particular 2023 peak happened to look attractive. Using the full-year periodogram to choose predictive frequencies would leak the evaluation period into the specification. The spectrum and the regression

therefore play different roles: the spectrum describes; the pre-specified Fourier basis forecasts.

Why stop at four daily and three weekly harmonics? The specification was deliberately kept small and written down before estimation. Adding a large candidate pool would introduce researcher degrees of freedom and turn the later confirmation into a test of a search procedure. The model asks a bounded question about temporal structure rather than trying to win a forecasting beauty contest.

The deliberately small specification produces a disappointing first result. On 2023 data the rolling Fourier/calendar model has MAE 558.8 MW, worse than D-7. But the error decomposition reveals something useful: its within-day shape error is already slightly better than persistence, while 87% of its error variance is day-level. It is often drawing roughly the right daily shape, but the entire predicted curve is vertically displaced - too high or too low.

2023 diagnostic	D-7	Fourier/calendar
Overall MAE (MW)	418.7	558.8
Day-level MAE (MW)	370.2	530.4
Within-day shape MAE (MW)	206.9	191.8
Level share of error variance	0.76	0.87

5. Fix the diagnosed problem, not the neighbouring columns

Notebook reference: §7, CELL 23-28 (recent-level correction, positive/negative leakage controls and information-matched D-7 control).

The diagnosis suggests one specific correction: estimate the recent level error that is already observable at the 18:00 origin and add the same scalar offset to every slot of the next-day structural forecast.

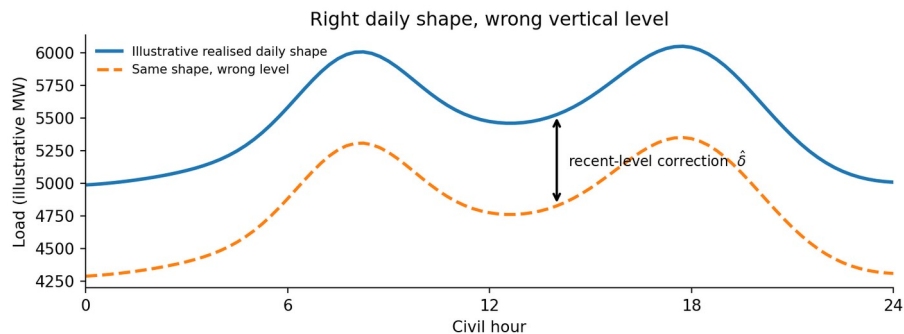


Illustration only: a scalar level correction shifts the whole target-day curve up or down without changing its intraday shape.

For each target day, the correction is the mean of historical out-of-sample residuals over the preceding 24 elapsed hours. Here out-of-sample means that each historical residual came from a forecast made without using that observation to fit itself. The correction is deliberately one scalar parameter, uses only information available by the origin, and preserves the model's intraday shape.

Positive and negative controls. If legitimate pre-origin residual-generating actuals are shifted, the correction must move. If actuals strictly after the origin are shifted by an absurd amount, the forecast must not move at all. Both behaviours are tested.

The 2023 development result is dramatic: MAE falls from 557.3 to 254.5 MW. That is not yet the scientific conclusion, because the correction itself might simply be a powerful recency feature.

So the same recent-level correction is also given to D-7. That creates an information-matched attribution benchmark. In the untouched 2025 confirmation, D-7 improves only from 438.9 to 423.0 MW, while Fourier/calendar plus the identical level information reaches 315.4 MW. The remaining gain is therefore associated with the structural forecast, not merely with access to recent load.

6. Freeze first, then open the future

Notebook reference: §10-§11, CELL 38-46 (probabilistic specification, pre-confirmation hardening, freeze record and opening of untouched 2025 confirmation).

After development and the 2024 point confirmation, the complete point and probabilistic specification is frozen in Git before any untouched 2025 confirmation result is scored. January 2025 had already been encountered during development, so it is declared burned - no longer eligible to count as untouched evidence - and excluded rather than quietly folded into the final sample.

The order of operations is part of the evidence



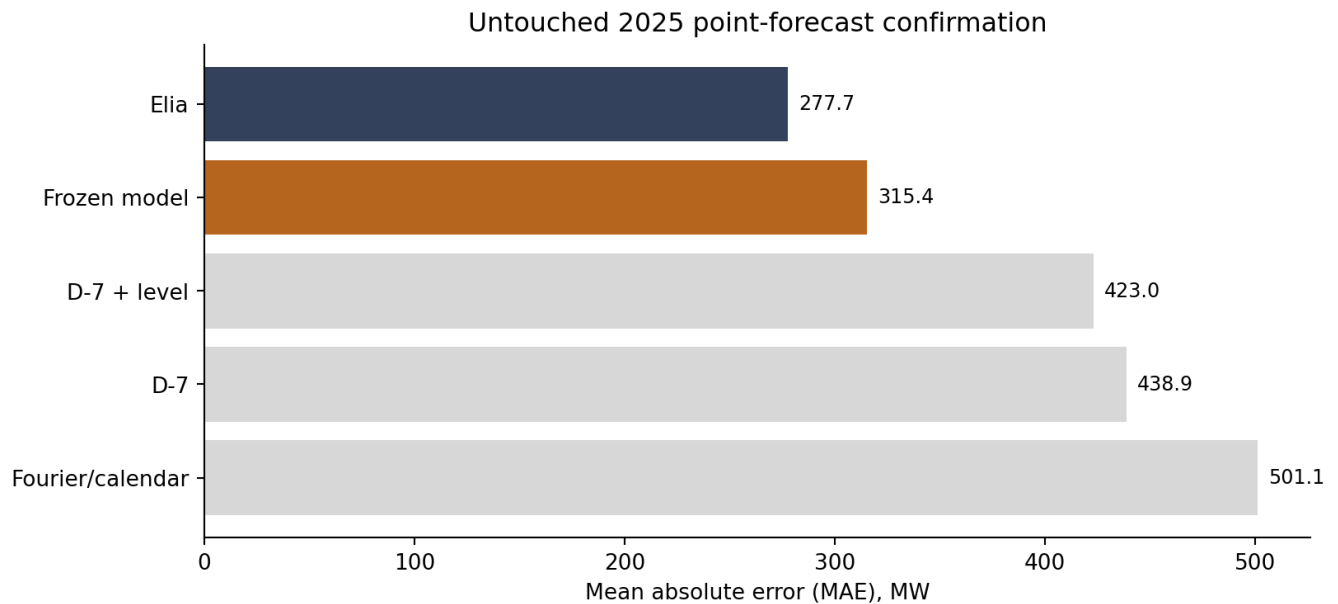
Confirmation period	Value
Scored dates	1 February-31 December 2025
Civil days	334
Quarter-hour slots	32,064
Point common sample	32,060
Reason for four exclusions	D-7 source times absent on spring-forward day

The freeze is documentary evidence, not an access-control mechanism. Its force comes from chronology: the commit exists before the confirmation outcomes are opened, and the notebook records the specification and burned month explicitly.

7. Untouched point confirmation: the attribution survives

Notebook reference: §11, CELL 47-50 (2025 frame, frozen point forecasts, information-boundary audit and first untouched point score); §12, CELL 55-57 (post-confirmatory point inference).

Forecast	MAE (MW)
Previous week D-7	438.9
D-7 + identical recent level	423.0
Fourier/calendar	501.1
Fourier/calendar + recent level	315.4
Elia day-ahead	277.7



The frozen model improves MAE by 25.4% relative to the information-matched D-7 + recent-level benchmark. A seven-day moving-block bootstrap - repeated resampling of consecutive seven-day groups to preserve short-range dependence - with 2,000 replicates gives a 95% interval of 18.8% to 32.0%. This is the clean positive result of the study.

The development result had been more optimistic: 254.5 MW versus 315.4 MW in untouched confirmation. That deterioration is not a defect to hide; it is exactly why the independent confirmation period exists.

Elia remains better: the frozen model's MAE is 13.6% higher, with a 95% block-bootstrap interval of 5.9% to 21.5%. That comparison is intentionally demanding. Elia's operational forecast may use information unavailable to a deliberately load-history-only model, so it is an operational reference rather than a like-for-like competitor.

What survived confirmation. Recurring load shape and recent level carry complementary predictive information. The raw Fourier/calendar model is poor; the same recent-level feature barely helps persistence; together, structure plus level produce a large confirmed gain.

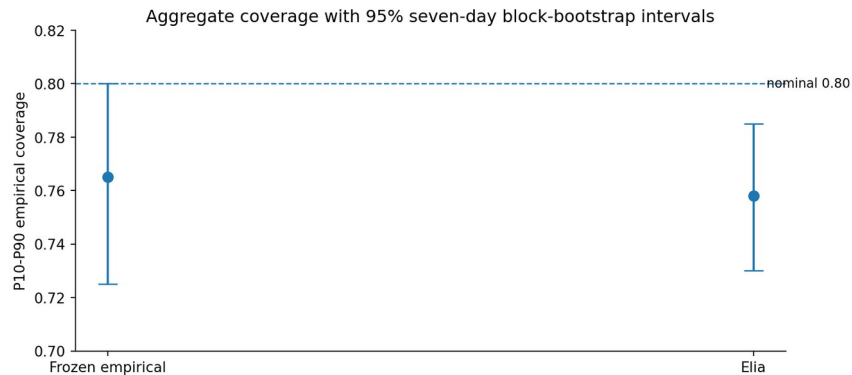
8. Probabilistic forecasting: do recent residuals quantify uncertainty?

Notebook reference: §10, CELL 38-43 (probabilistic specification and 2024 development diagnostics); §11, CELL 51-52 (untouched 2025 probabilistic confirmation); §12, CELL 58 (probabilistic inference).

The point forecast is only part of an operational decision. The study therefore adds a deliberately simple uncertainty model. For each target quarter-hour, it looks back 56 elapsed days, takes historical residuals from the same Belgian civil hour, and adds their empirical 10th and 90th percentiles to the frozen point forecast. Coverage is the fraction of realised loads that fall inside this P10-P90 band.

No parametric error distribution is assumed. There is no width tuning, calibration scaling or hyperparameter search. The same origin guard applies to the residual pool. Pinball loss is the standard loss for a quantile forecast: it penalises misses asymmetrically according to the requested quantile, and lower is better.

Forecast	Coverage	Mean width (MW)	Mean pinball (MW)
Frozen empirical residuals	0.765	941.9	74.19
Elia day-ahead	0.758	836.1	66.09
Nominal	0.800	-	-



The observed coverage difference is small relative to dependence-aware uncertainty; the nominal 0.80 level lies at the upper edge of the frozen-model interval.

The frozen-model coverage interval is 0.725-0.800, so this confirmation does not establish departure from nominal 80% coverage. The paired coverage difference versus Elia is +0.007 with interval -0.033 to +0.042: no aggregate coverage advantage is established.

The frozen quantile forecasts are 12.7% wider and have 12.3% higher mean pinball loss than Elia. But those differences cannot be assigned to the interval construction alone: the underlying frozen point forecast already has 13.6% higher MAE. The experiment does not identify how much of the probabilistic gap is centering and how much is interval construction.

9. Conditional calibration: useful diagnostics, restrained claims

Notebook reference: §11, CELL 53 (untouched conditional calibration); §12, CELL 59 (dependence-aware uncertainty for the pre-specified conditional cells).

Season	Frozen coverage	95% block-bootstrap interval
DJF*	0.824	0.728-0.899
MAM	0.779	0.708-0.852
JJA	0.792	0.734-0.848
SON	0.684	0.611-0.746

* DJF contains February and December only because January 2025 was burned before confirmation.

SON is the only pre-specified seasonal/day-type cell whose ordinary 95% block-bootstrap interval excludes nominal 0.80 coverage. It is treated as a conditional diagnostic, not as a multiplicity-adjusted confirmatory finding: once several conditional cells are inspected, the evidence for any one extreme cell should be interpreted more cautiously than if it had been the only hypothesis tested. The holiday estimate is based on only nine civil days and is too imprecise for a substantive holiday-effect claim.

10. Why the bootstrap resamples days, not quarter-hours

Notebook reference: §12, CELL 55-60 (post-confirmatory reporting and seven-day moving-block bootstrap).

There are more than 32,000 quarter-hour observations in the confirmation period, but they are not 32,000 independent experiments. Errors within a day are strongly related, and neighbouring days can share weather, calendar and level conditions.

The post-confirmatory uncertainty analysis therefore uses a seven-day moving-block bootstrap over complete Belgian civil days, with 2,000 replicates. Within each replicate the same sampled days are used for every model, so differences and ratios remain paired. The 92- and 100-slot daylight-saving days retain their natural slot weights.

Headline comparison	Estimate	95% moving-block interval
Frozen vs D-7 + identical level	25.4% lower MAE	18.8% to 32.0% lower
Frozen vs plain D-7	28.1% lower MAE	20.1% to 35.7% lower
Frozen vs Elia	13.6% higher MAE	5.9% to 21.5% higher
Coverage difference vs Elia	+0.7 percentage points	-3.3 to +4.2 pp
Interval width vs Elia	12.7% higher	5.8% to 20.1% higher
Mean pinball vs Elia	12.3% higher	3.5% to 22.4% higher

The bootstrap is non-circular, so observations near the beginning and end of the period occur in fewer candidate blocks than interior days. The study records this small boundary effect and does not rerun the analysis post hoc to remove it.

11. What did we actually learn?

1. Periodicity is not predictability. The realised series has strong daily and weekly spectral structure, yet the raw Fourier/calendar forecast loses badly to weekly persistence.

2. Shape and level are different forecasting problems. The structural model captures within-day shape reasonably well but misses day-level positioning. A single leakage-safe recent-level correction fixes the diagnosed deficiency.

3. Attribution needs an information-matched control. Giving the same recent-level correction to D-7 shows that recency alone does not explain the 315.4 MW result.

4. Coverage is competitive, but the intervals are less efficient. Aggregate frozen-model coverage is not distinguishable from nominal 80% coverage or from Elia, while the complete frozen quantile forecast is wider and has higher pinball loss. The study stops where causal attribution stops.

5. Experimental discipline is part of the product. The specification is frozen before untouched outcomes are opened; a burned month is declared; leakage controls work in both directions; common samples are enforced; and dependence-aware intervals accompany the headline comparisons.

This is not presented as a new state-of-the-art load forecaster. It is a worked example of building a forecasting experiment whose claims can be traced back to a defined information set, a frozen specification, an attribution control and an untouched confirmation.

12. From forecast accuracy to decisions

The natural next question is not whether another feature can shave a few megawatts from MAE. It is whether statistical forecast accuracy translates into economic value when a constrained battery must decide when to charge and discharge.

That relationship need not be monotone. A forecast error in a flat-price hour may cost almost nothing; a smaller error near a price spread or state-of-charge boundary may change the dispatch materially. A 13.6% MAE gap therefore need not become a 13.6% euro gap.

Next research question. Does improved statistical forecast accuracy translate into improved economic battery dispatch - and under what price and state-of-charge conditions does the ranking of forecasts change?

The information-boundary discipline transfers directly. A dispatch policy must be optimized against information available at decision time. Realised prices may be used afterwards to value the dispatch, but optimizing against realised future prices and then reporting the resulting value would be perfect-foresight leakage.

Glossary

Term	Meaning
D-7 persistence	Forecast using load at the same Belgian civil time one week earlier.
Civil time	Europe/Brussels clock time, including daylight-saving transitions.
Fourier harmonic	Sine/cosine component representing a fixed periodicity such as 24 or 12 hours.
Recent-level correction	Single additive offset estimated from forecast residuals observed in the preceding 24 elapsed hours.
P10 / P90	10th and 90th predictive quantiles; together they form a nominal 80% interval.
Pinball loss	Loss function for quantile forecasts. It penalises forecasts differently depending on the requested quantile; lower is better.
Coverage	Fraction of realised loads that fall inside the forecast interval; for a P10-P90 band the nominal target is 80%.
Moving-block bootstrap	Resampling consecutive groups of days to preserve short-range dependence.
Untouched confirmation	Evaluation period whose outcomes were not used to choose or tune the frozen specification.
Information-matched control	Comparator given the same additional information so the incremental contribution of model structure can be isolated.
MAE	Mean absolute error: the average absolute difference between realised and forecast load, measured here in MW. Lower is better.
Leakage	Use of information that would not have been available when the forecast was issued. Leakage can make a model look better in evaluation than it could have been in operation.
Forecast origin	The time at which a forecast is treated as issued. Here it is 18:00 Europe/Brussels on D-1, matching the ODS001 field metadata.
Information boundary	The cutoff implied by the forecast origin: only observations known by that time may influence the forecast or its uncertainty interval.

Source and status note

Elia data were retrieved and cached during notebook execution; the exact acquisition timestamp was not recorded, and later historical pulls may differ slightly if Elia revises observations. No new empirical result has been invented for this guided tour.

Headline evidence: 334 confirmation days; 32,060 common point slots; frozen-model MAE 315.4 MW versus 423.0 MW for the information-matched D-7 + recent-level control and 277.7 MW for Elia; paired improvement 25.4% with 95% seven-day moving-block interval 18.8%-32.0%; frozen P10-P90 coverage 0.765 with interval 0.725-0.800. The bootstrap used 2,000 replicates.

This booklet summarises the ElectricityLoadForecasting repository, computational notebook and tested Python package. The complete point and probabilistic specification was frozen in Git before the February-December 2025 confirmation outcomes were scored; January 2025 had already been seen and is excluded from untouched claims.